## 1. Differential Equations

- Form a differential equation using the information given
  - If something is proportional, add constant of proportionality $k$
  - If rate is decreasing, add a negative sign
- Separate variables, bring $\mathrm{dx}$ and $\mathrm{dt}$ on opposite sides
  - Integrate both sides to form an equation
  - Add arbitrary constant
  - Use conditions given to find $c$ and/or $k$

### {W10-P33} Question 9:

A biologist is investigating the spread of a weed in a particular region. At time $t$ weeks, the area covered by the weed is $Am^2$ . The biologist claims that rate of increase of $A$ is proportional to $\sqrt{2\mathrm{A}-5}$.

1. Write down a differential equation given info
2. At start of investigation, area covered by weed was $7m^2$ . $10$ weeks later, area covered $= 27m^2$ Find the area covered $20$ weeks after the start of the investigation.

**Solution:**

**Part (i)**

$$\frac{\mathrm{dA}}{\mathrm{dt}} \propto \sqrt{2A-5} = k\sqrt{2A-5}$$

**Part (ii)**

*Proceed to form an equation in* $\mathrm{A}$ *and* $\mathrm{t}$ *:*

$$\frac{\mathrm{dA}}{\mathrm{dt}} = k\sqrt{2A-5}$$

*Separate variables*

$$\frac{1}{\sqrt{2A-5}}dA = kdt$$

*Integrate both sides*

$$kt + c = (2A-5)^{\frac{1}{2}}$$

*When* $t = 0$ *:*

$A = 7 \therefore c = 3$

$$kt + 3 = (2A-5)^{\frac{1}{2}}$$

*When* $t = 10$ *:*

$$10k + 3 = (2(27)-5)^{\frac{1}{2}}$$

$$10k = \sqrt{49} - 3$$

$$k = 0.4$$

*Now substitute* $20$ *as* $\mathrm{t}$ *and then find* $\mathrm{A}$ *:*

$$0.4(20) + 3 = (2A-5)^{\frac{1}{2}}$$

$$11 = (2A-5)^{\frac{1}{2}}$$

$$121 = 2A - 5$$

$$A = 63m^2$$

### {S13-P31} Question 10:

Liquid is flowing into a small tank which has a leak. Initially the tank is empty and, $t$ minutes later, the volume of liquid in the tank is $\mathrm{V}\ cm^3$ . The liquid is flowing into the tank at a constant rate of $80\ cm^3$ per minute. Because of the leak, liquid is being lost from the tank at a rate which, at any instant, is equal to $\mathrm{kV}\ cm^3$ per minute where $\mathrm{k}$ is a positive constant.

i. Write down a differential equation describing this situation and solve it to show that:

$$V = \frac{1}{k}\left(80 - 80e^{-kt}\right)$$

$$k = \frac{4 - 4e^{-15k}}{25}$$

ii. Work out the volume of liquid at $t = 20$ and state what happens to volume after a long time.

**Solution:**

**Part (i)**

*Represent the given information as a derivative:*

$$\frac{\mathrm{dV}}{\mathrm{dt}} = 80 - kV$$

*Proceed to solve the differential equation:*

$$\frac{\mathrm{dt}}{\mathrm{dV}} = \frac{1}{80 - kV}$$

$$dt = \frac{1}{80 - kV}\mathrm{dV}$$

$$\int (1)\,\mathrm{dt} = \int \frac{1}{80 - kV}\mathrm{dV}$$

$$t + c = -\frac{1}{k}\ln|80 - kV|$$

*Use the given information; when* $t = 0,\ V = 0$ *:*

$$\therefore c = -\frac{1}{k}\ln(80)$$

*Substitute back into equation:*

$$t - \frac{1}{k}\ln(80) = -\frac{1}{k}\ln|80 - kV|$$

$$t = \frac{1}{k}\ln(80) - \frac{1}{k}\ln|80 - kV|$$



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$$t = \frac{1}{k}\ln\left(\frac{80}{80-kV}\right)$$

$$kt = \ln\left(\frac{80}{80-kV}\right)$$

$$e^{\mathrm{kt}} = \frac{80}{80-kV}$$

$$80 - kV = \frac{80}{e^{\mathrm{kt}}}$$

$$kV = 80 - 80e^{-kt}$$

$$V = \frac{1}{k}\left(80 - 80e^{-kt}\right)$$

**Part (ii)**

*After carrying out the iterations, the following result will be obtained:*

$$k = 0.14\ \ (2d.p.)$$

**Part (iii)**

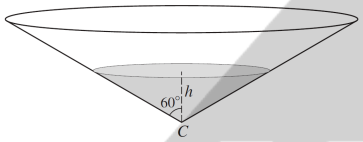
*Simply substitute into the equation's* $\mathrm{t}$*:*

$$V = \frac{1}{0.14}\left(80 - 80e^{-0.14(20)}\right) = 537\ cm^3$$

*The volume of liquid in the tank after a long time* ***approaches*** *the max volume:*

$$V = \frac{1}{0.14}(80) = 571\ cm^3$$

**{W13-P31} Question 10:**




A tank containing water is in the form of a cone with vertex $\mathrm{C}$. The axis is vertical and the semi-vertical angle is $60^\circ$, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is $H$. At this instant, a tap at $C$ is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where $\mathrm{h}$ is the depth of water at time *t*. The tank becomes empty when $t = 60$.

$$\frac{\mathrm{dh}}{\mathrm{dt}} = -Ah^{-\frac{3}{2}}$$

i. Solve differential equation given in part i and obtain an expression for $\mathrm{t}$ in terms of $\mathrm{h}$ and $\mathrm{H}$.

**Solution:**

**Part (i)**

*First represent info they give us as an equation:*

$$V = \frac{1}{3}\pi r^2 h$$

$$r = \tan 60 \times h = h\sqrt{3}$$

$$\therefore V = \frac{1}{3}\pi\left(h\sqrt{3}\right)^2 h = \pi h^3$$

$$\frac{\mathrm{dV}}{\mathrm{dh}} = 3\pi h^2$$

$$\frac{\mathrm{dV}}{\mathrm{dt}} \propto -\sqrt{h} = -kh^{\frac{1}{2}}$$

*Find the rate of change of* $\mathrm{h}$*:*

$$\frac{\mathrm{dh}}{\mathrm{dt}} = \frac{\mathrm{dV}}{\mathrm{dt}} \div \frac{\mathrm{dV}}{\mathrm{dh}}$$

$$\frac{\mathrm{dh}}{\mathrm{dt}} = \frac{-kh^{\frac{1}{2}}}{3\pi h^2} = -\frac{k}{3\pi}h^{-\frac{3}{2}}$$

**Part (ii)**

$$dt = \frac{1}{-Ah^{-\frac{3}{2}}}\mathrm{dh}$$

$$\int \mathrm{Adt} = \int \frac{1}{-h^{-\frac{3}{2}}}\mathrm{dh}$$

$$At + c = -\frac{2}{5}h^{\frac{5}{2}}$$

*Use given information to find unknowns; when* $t = 0$ *:*

$$-A(0) + c = \frac{2}{5}(H)^{\frac{5}{2}} \quad \therefore c = \frac{2}{5}H^{\frac{5}{2}}$$

*When* $t = 60$ *:*

$$-A(60) + c = 0$$

$$c = 60A$$

$$A = \frac{1}{150}H^{\frac{5}{2}}$$

*Thus the initial equation becomes:*

$$-\frac{1}{150}H^{\frac{5}{2}}t + \frac{2}{5}H^{\frac{5}{2}} = \frac{2}{5}h^{\frac{5}{2}}$$

$$H^{\frac{5}{2}}\left(-\frac{t}{150} + \frac{2}{5}\right) = \frac{2}{5}h^{\frac{5}{2}}$$

$$-\frac{t}{150} + \frac{2}{5} = \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}$$

$$\frac{t}{150} = \frac{2}{5} - \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}$$

$$t = 150\left(\frac{2}{5} - \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}\right) = 60 - 60h^{\frac{5}{2}}H^{-\frac{5}{2}}$$

$$t = 60\left(1 - \left(\frac{h}{H}\right)^{\frac{5}{2}}\right)$$



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